

Insight into CO₂/CH₄ separation performance in ionic liquids/polymer membrane from molecular dynamics simulation



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ABSTRACT

Owing to the advantages of both ionic liquids (ILs) and polymer membranes, ILs/polymer hybrid membranes are becoming one of the research hotspots in the CO₂ separation. However, the lack of molecular insight into the CO₂ separation mechanism restricts the development of ILs/polymer hybrid membranes. Herein, the separation process of CO₂/CH₄ through the membrane composed by poly(vinylidene fluoride) (PVDF) with 1-butyl-3-methylimidazolium hexafluorophosphate ([Bmim][PF₆]) and 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([Bmim][NTf₂]) were studied by molecular dynamics simulations. Structure analysis indicates that PVDF chains tend to aggregate on the surface with ILs being confined inside the membrane, but the ILs within the interfacial regions can dramatically weaken the crystallinity of PVDF and improve the flexibility of PVDF chains, thus benefiting the gas permeation and diffusion. In addition, the calculated results of the permeation selectivity, the solubility selectivity and the diffusion selectivity confirm that [Bmim][NTf₂]/PVDF membrane shows better performance in CO₂/CH₄ separation than [Bmim][PF₆]/PVDF membrane, and the solubility selectivity plays a dominant role. The high selectivity in the hybrid membranes is due to the stronger interaction between CO₂ and ILs/PVDF compared to that for CH₄. Detailed analysis on the CO₂ transport process suggests that the driving force for CO₂ penetrating onto the interface is about $-4.11 \text{ kJ} \cdot \text{mol}^{-1}$, and most of CO₂ at the interface are nearly parallel to the membrane. Moreover, the selectivity of CO₂/CH₄ increases as the thickness of membranes decreases.

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1. Introduction

With the increasing demand for energy and environmental protection in the world, the development and utilization of gas clean energy including natural gas [1], shale gas [2,3], biomethane [4], and so on, have become a major trend. However, in addition to the main component CH₄, these gases also contain CO₂. The presence of CO₂ not only reduces the heat capacity and quality, increases the cost of transportation [5] and storage [6], but also corrodes equipment and pipelines [7]. Therefore, efficient separation of CO₂ [8] before transportation and utilization is of great importance.

Membranes have gained widely attention for separating gases due to low energy consumption, simple equipment, easy operation

and low investment [9,10]. For the past 20 years, a lot of fluoropolymer membrane materials have been developed to improve the separation efficiency of CO₂ [11–17].

Recently, ionic liquids (ILs) [18–22] have been regarded as potential CO₂ absorbent due to low volatility, high thermal stability, high solubility and selectivity of CO₂ compared to other gases, such as CH₄, N₂ [23–26]. Up to now, a lot of ILs were synthesized for separating CO₂ [27–30]. However, ILs directly used as solvents for CO₂ separation have some problems that cannot be ignored, such as high viscosity and cost [31–33]. Therefore, some researchers have combined ILs and fluoropolymer to make membrane to separate CO₂ [17,34–37], which not only can effectively avoid the high viscosity problem caused by using ILs directly, but also can reduce the amount and loss of ILs. For example, Lee et al. [38] prepared 1-butyl-3-methylimidazolium tetrafluoroborate ([Bmim][BF₄])–PVDF blended membrane for separating of CO₂ from natural gas with the CO₂/CH₄ selectivity up to 50–100. Swati et al. [39] found that the CO₂/CH₄ selectivity in 1-butylsulfonate–3-methyl

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midazolium P-toluene sulfonate ([Bsmim][tos]) and 1-butylsulfonate pyridine P-toluene sulfonate ([BsmPy][tos])-PVDF composite membranes were 47.4 and 40.7, respectively.

Although a lot of experimental studies have confirmed that composite membranes composed by fluoropolymers and ILs can increase the CO₂ permeability and selectivity. However, the mechanism of CO₂ in composite membranes is still indistinct, which is important information in for developing new membranes. Molecular dynamics (MD) simulation is an effective method to study microscopic mechanism [40–45]. For example, the effect of nano-constraint on the CO₂/CH₄ separation in a PVDF/IL was studied by Rahmani et al. through combining experiment and MD simulation. They found that the CO₂/CH₄ selectivity increased with decreasing nano-constraint [46]. Our previous MD simulation results [47] revealed that with the increase of IL content, a continuous IL channel gradually formed inside the composite membrane. Szala-Bilnik et al. [48] simulated and reported that the solubility and diffusivity of CO₂ increased with the addition of ILs in the polymer membrane.

Nevertheless, the current studies on the theory of CO₂ in ILs/polymer hybrid membranes is limited and confined to the bulk phase. To the best of our knowledge, there are no studies focus on the interface for the system of CO₂/CH₄ in ILs/polymer hybrid membranes until now. In this work, the interface of CO₂/CH₄ in ILs/PVDF hybrid membranes with different thicknesses were simulated and the solubility-diffusion mechanism were investigated by MD simulation. The two typical ILs [Bmim][PF₆] and [Bmim][NTf₂] with high CO₂ solubility and low CH₄ solubility were selected. The thickness of 3, 4, 5 and 6 nm were considered. The structural morphology of the ILs/PVDF membranes and the separation performance of CO₂/CH₄ were analyzed. The influence of ILs on the separation of CO₂/CH₄ was studied by analyzing the interaction and kinetic properties. Then, the behavior of CO₂ at the interface of the hybrid membrane and the average force potential of CO₂ into the membrane were analyzed. Finally, the effect of the thickness of membrane were investigated by calculating the permselectivity, radial distribution function and diffusion coefficient.

2. Methodology

All MD simulations were performed by Gromacs package (v.5.1.4) [49,50]. The all-atom model was adopted to simulate the system of CO₂/CH₄ in [Bmim][PF₆]/[Bmim][NTf₂] and PVDF. The force field parameters for [Bmim]⁺ and [PF₆][−] developed by Liu et al. [51] were adopted. The force field parameters for [NTf₂][−] was taken from Canongia Lopes et al. [52]. The parameters for PVDF, CO₂ and CH₄ were procured from Lachet et al. [53], Shi et al. [54] and Skoulidas et al. [55], respectively. Fig. 1 shows the molecular structures of [Bmim]⁺, [PF₆][−], [NTf₂][−] and PVDF.

The whole simulation process was divided into two stages. The first stage was the simulation of the hybrid membrane of ILs and

Table 1

Simulation systems of the hybrid ILs/PVDF and pure PVDF membranes.

membrane	Thickness	number of PVDF chains	number of ILs
ILs/PVDF membrane	3 nm	75	175
	4 nm	100	233
	5 nm	125	291
	6 nm	150	349
pure PVDF membrane	6 nm	300	0

PVDF. According to our previous work, PVDF was represented by several individual chains made up of 10 monomers (i.e., 20 carbon atoms) [47]. The initial configurations were prepared using Packmol software to randomly pack ILs and PVDF chains in a cubic cell at a weight ratio of 1:1, which were set based on the experiment. For example, Chen et al. [56] and Hong et al. [57] successfully synthesized PVDF/[emim][B(CN)₄] and [emim][BF₄]/PVDF-HFP blend membranes with the weight ratio of 1:1, respectively. The numbers of PVDF chains and ILs are given in Table 1. The prepared hybrid membrane was then equilibrated in an isothermal-isobaric ensemble (NPT) with periodic boundary conditions in the xyz directions for 100 ns with a 2 fs time step. All simulations were performed at 493 K and 1 bar. Fig. 2(a–d) shows the hybrid membranes with dimensions of approximately 65 Å by 65 Å in the x and y directions, respectively. The membranes of different thicknesses (3, 4, 5 and 6 nm) were set by adjusting the amount of ILs and PVDF (see Table 1).

The second stage was the simulation of the system with CO₂/CH₄ and ILs/PVDF membranes. As shown in Fig. 2e, the ILs/PVDF hybrid membrane were fixed at the center of a box, and 100 gas molecules were placed in the gas phase above the hybrid membrane on the feed-side in the three-chamber system. The initial height of the feed-side chamber was set by the Peng-Robinson equation to ensure 10 bar as the initial air pressure. To be close to the real state, the PVDF was fixed in the z-direction. In the three-chamber box, a vacuum layer was added on the other side of the hybrid membrane as the permeate-side chamber, which has the same height as the feed-side chamber. The periodic boundary condition was applied in the x and y directions, and two virtual walls were added on the z-direction. All simulations were ran for 140 ns in a canonical ensemble (NVT) with a 1 fs time step and a temperature of 298 K.

3. Results and discussion

3.1. Effect of ILs on membrane

It is well known that the structure of the hybrid membrane has a great influence on the efficiency of gas separation. Therefore, the effect of ILs on the microstructure of the hybrid membrane was investigated at first.

As an example, the snapshot of the pure PVDF membrane and the ILs/PVDF membrane are shown in Fig. 3. Fig. 3a clearly shows that the extensive crystalline state exist in the pure PVDF membrane, while Fig. 3(b–c) indicates that in the hybrid membrane there are no obvious crystallization, which agrees with the recent experimental report [58]. The reduction of PVDF crystallization was suggested to be contributed from the destroyed hydrogen bonding between the PVDF with the ILs [47].

The effect of ILs on the polymer structure is also confirmed by the center-of-mass radial distribution functions (RDFs) of PVDF chains in different membranes shown in Fig. 4a. It is clear that the first peak intensity decreases significantly under the accretion

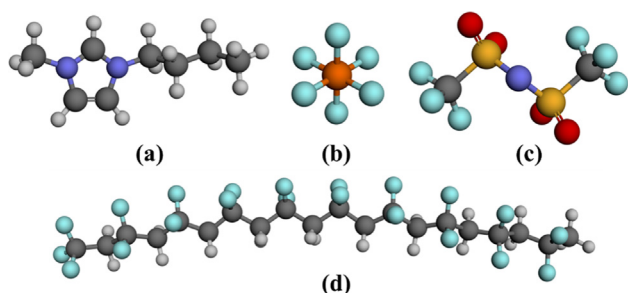


Fig. 1. The atomic structure of (a) [Bmim]⁺, (b) [PF₆][−], (c) [NTf₂][−] and (d) PVDF chain.

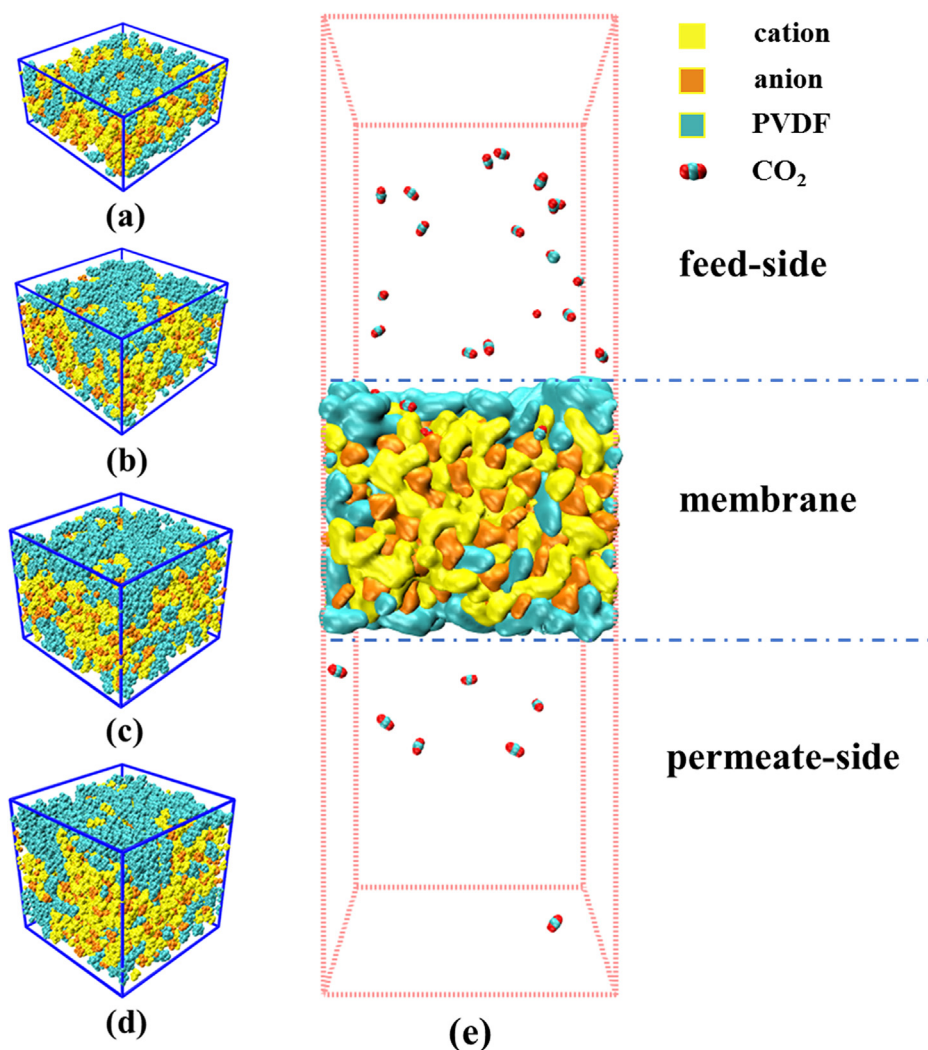


Fig. 2. (a–d) The hybrid membranes of different thicknesses (3, 4, 5 and 6 nm). (e) Three-chamber system used to simulate the gas permeation through the hybrid membrane.

of ILs, indicating that the interaction between PVDF is largely reduced. The presence of multiple peaks in the RDF of the pure PVDF membrane indicates the presence of multiple coordination shells between PVDF, corresponding to the high crystallization of PVDF. However, there are only three peaks in the RDF of the [Bmim][PF₆]/PVDF system, and only one distinct peak in the RDF of the [Bmim][NTf₂]/PVDF system, further supporting that the addition of ILs can weaken the crystalline structure of PVDF and improve the flexibility of PVDF chain segments. Moreover, for the studied two kinds of ILs, the first peak value for PVDF chains in the [Bmim][NTf₂]/PVDF membrane is lower than that in the [Bmim][PF₆]/PVDF membrane, suggesting that the interaction between PVDF chains and [Bmim][NTf₂] is weaker than that with [Bmim][PF₆].

The significant change on the membrane structure leads to different kinetic properties of PVDF chains in these membranes. Fig. 4b gives the mean square displacement (MSD) of the PVDF chains. The MSDs of PVDF in two ILs/PVDF hybrid membranes are significantly larger than that in pure PVDF, suggesting that PVDF in the two ILs/PVDF hybrid membranes diffuse faster than that in the pure PVDF membrane. Moreover, the MSD of PVDF in [Bmim][NTf₂]/PVDF is larger than that in [Bmim][PF₆]/PVDF, indicating that the diffusion of PVDF in the former is faster than that in the later.

In general, ILs can reduce the crystallization of PVDF in the hybrid membrane, making the polymer chains more flexible and diffusible, which, eventually, can facilitate the gas permeation and diffusion through the hybrid membrane as we show below.

3.2. Distribution of ILs and PVDF

To clarify the overall structures of different membranes, the number density of PVDF, cations and anions in the pure PVDF and ILs/PVDF hybrid membranes were calculated and are given in Fig. 5. The two surfaces of the membrane are roughly at $z = -3$ and 3 nm, respectively. As shown in Fig. 5a, for the pure PVDF membrane, the number density distribution of PVDF is very uniform on the interface of the membrane. However, the density of PVDF inside the membrane is small and uneven. In combination with Fig. 3a, it can be inferred that PVDF crystallizes on the membrane surface.

In contrast, as shown in Fig. 5(b–c), in the ILs/PVDF membranes, the [Bmim]⁺, [PF₆][−] and [NTf₂][−] are mainly concentrated inside the hybrid membrane. A small amount of [Bmim]⁺, [PF₆][−] and [NTf₂][−] are in the interfacial regions of the hybrid membrane. Meanwhile, the PVDF aggregates on the surface of both sides of the hybrid membrane. This sandwich-like structure can facilitate the fixation of ILs inside the hybrid membrane and avoids the loss of ILs. Chen

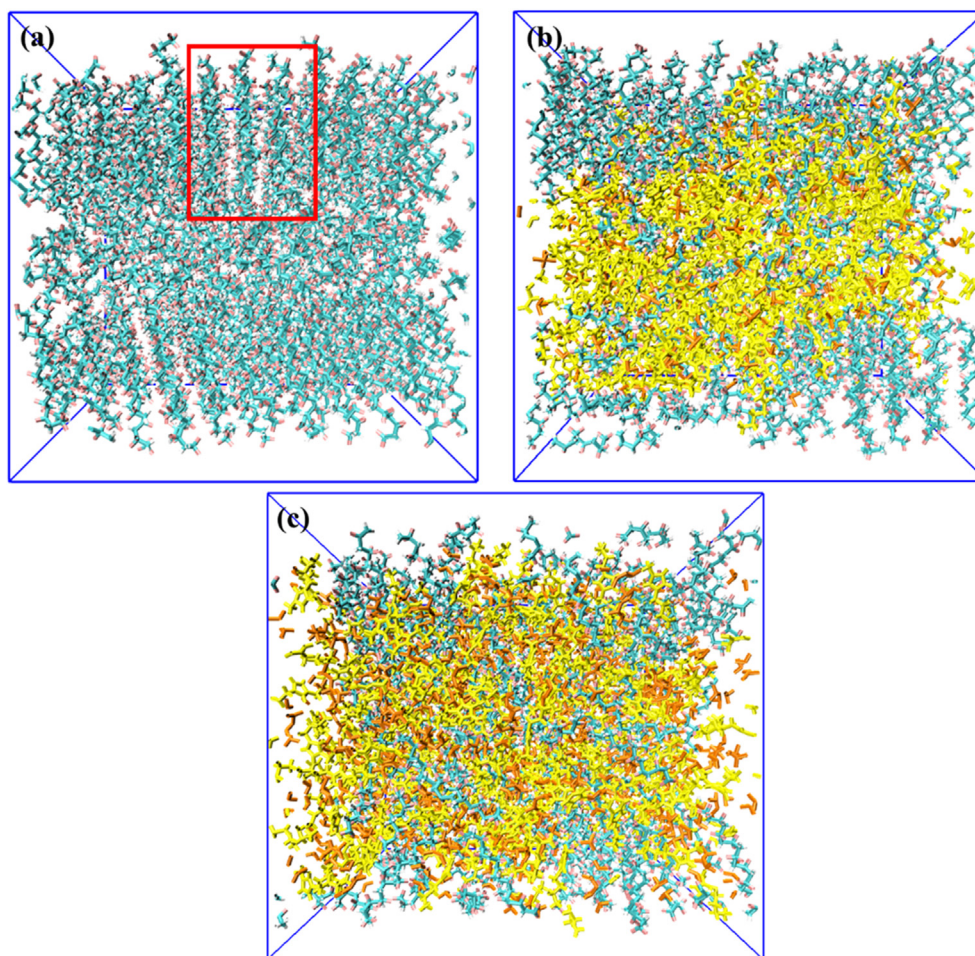


Fig. 3. Typical snapshots of (a) the pure PVDF, (b) the [Bmim][PF₆]/PVDF membranes and [Bmim][NTf₂]/PVDF membranes.

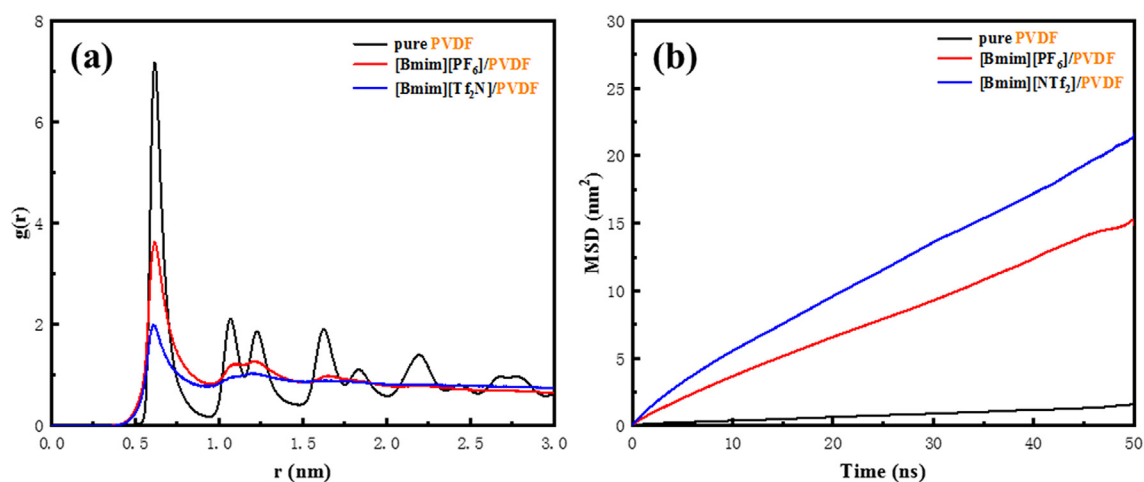


Fig. 4. (a) The center-of-mass RDFs between PVDF chains and (b) MSD vs. time for PVDF chains in the pure PVDF and the ILs/PVDF membranes.

et al. [56] also reported that both the ILs and crystallization of PVDF distribute at the membrane interface by experiment. However, the PVDF is not uniformly distributed at the interface of hybrid membrane, which maybe due to the reduction crystallization of PVDF caused by the ILs. The same number density profiles were also found for the thicknesses of 3, 4 and 5 nm, which are shown in Figure S1.

Moreover, from Fig. 5(b–c), it can be found that the peak height of PVDF for the [Bmim][NTf₂]/PVDF system is lower than that for the [Bmim][PF₆]/PVDF system, indicating that the crystallization of PVDF in [Bmim][NTf₂]/PVDF is less than that in [Bmim][PF₆]/PVDF. Furthermore, the distribution of [Bmim] cations and [NTf₂][−] are more homogeneous than those of [Bmim]⁺ and [PF₆][−]. Thus, compared to [Bmim][PF₆], we expect that the [Bmim][NTf₂] system

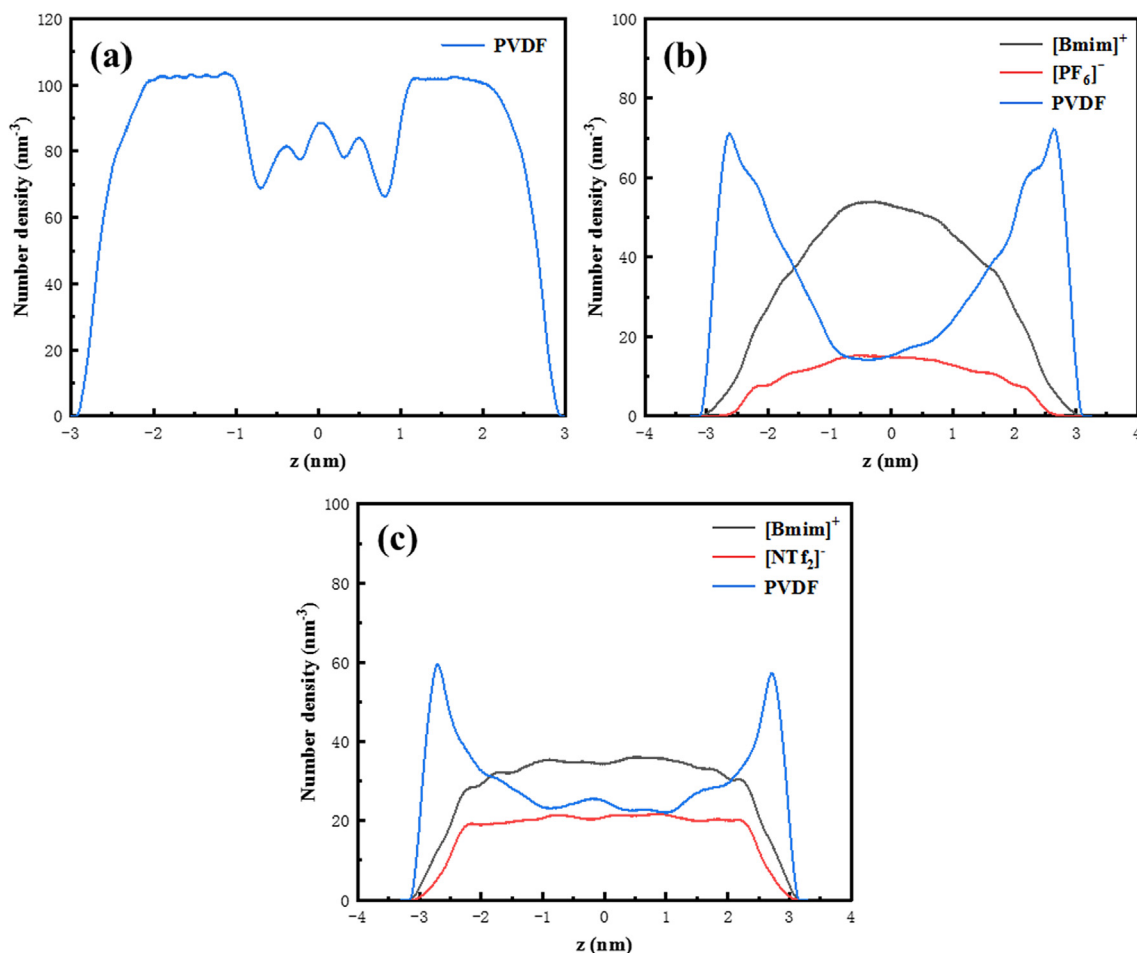


Fig. 5. The number densities of PVDF chains, ILs in (a) the pure PVDF, (b) the [Bmim][PF₆]/PVDF and (c) the [Bmim][NTf₂]/PVDF membranes.

show better separation performance of gas in ILs/PVDF hybrid membrane, which is reported in previous experiment [56].

3.3. Separation performance

To verify the separation performance of the [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF hybrid membranes, the solubility and permeation of CO₂ and CH₄ in these membranes were calculated and displayed in Fig. 6. From Fig. 6, it can be seen that both the [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF hybrid membranes are highly selective for CO₂, while those for CH₄ are less. Based on the results of Fig. 6, the selectivity CO₂/CH₄ in ILs/PVDF hybrid membranes can be obtained, which are displayed in Table 2. For [Bmim][PF₆]/PVDF membrane, the permeation selectivity, solubility selectivity and diffusion selectivity for CO₂/CH₄ are 2.24, 2.15 and 1.04, respectively. Whereas, for [Bmim][NTf₂]/PVDF membrane, the permeation selectivity, solubility selectivity, and diffusion selectivity are 2.93, 2.74, and 1.07 for CO₂/CH₄, respectively. It can be found that the permeation selectivity, solubility selectivity, and diffusion selectivity of CO₂/CH₄ in [Bmim][NTf₂]/PVDF membrane are higher than those in [Bmim][PF₆]/PVDF membrane. Therefore, [Bmim][NTf₂]/PVDF membrane has better separation performance in separating CO₂ and CH₄. Similarly, Jindratsamee et al. [59] reported that [Bmim][NTf₂]/PVDF membrane are more selective for CO₂/N₂ than [Bmim][PF₆]/PVDF membrane by experiment. In addition, the solubility selectivity is higher than diffusion selectivity both for [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF hybrid membranes,

revealing that the solubility selectivity plays a dominant role in the separation of CO₂/CH₄.

Moreover, the number densities of CO₂ and CH₄ in pure PVDF membrane were calculated and shown in Figure S3. From Figure S3, it can be found that only a small amount of CO₂ and almost no CH₄ soluble into the pure PVDF membrane, and CO₂ cannot penetrate into the permeate side. Chen et al. also demonstrated that the solubility of CO₂ in pure PVDF membranes is very low and less than that in ILs/PVDF hybrid membranes [56]. Therefore, compared with pure PVDF membrane, the selectivity of CO₂/CH₄ in two ILs/PVDF hybrid membranes is enhanced by adding ILs.

3.4. Gas-membrane interaction

In order to investigate the interaction between CO₂/CH₄ and ILs/PVDF hybrid membrane, the center-of-mass RDFs of gas-cations, gas-anions and gas-PVDF were calculated and depicted in Fig. 7. It is clear that the first peak values for CO₂-[Bmim]⁺, CO₂-[PF₆]⁻/[NTf₂]⁻ and CO₂-PVDF are higher than those for CH₄-[Bmim]⁺, CH₄-[PF₆]⁻/[NTf₂]⁻ and CH₄-PVDF both for the system of [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF, suggesting that the interaction of CO₂-[Bmim]⁺, CO₂-[PF₆]⁻/[NTf₂]⁻ and CO₂-PVDF are stronger than those for CH₄. Furthermore, the coordination numbers for CO₂/CH₄ around anions and PVDF at the first coordination sphere were analyzed and displayed in Table 3. As shown in Table 3, the coordination numbers of CO₂ are about 3 times higher than those of CH₄ for [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF hybrid membranes. It is further confirmed that the interaction

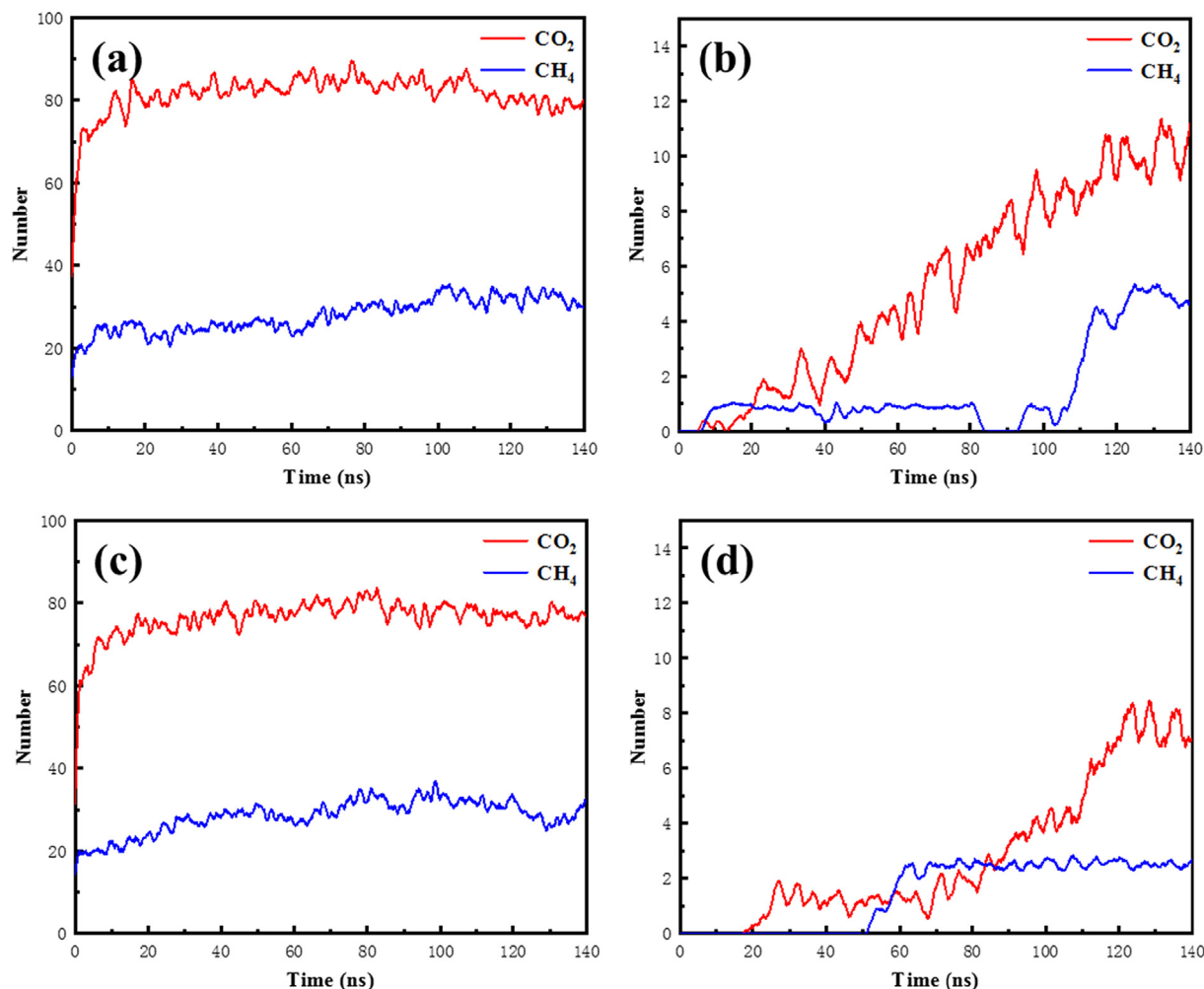


Fig. 6. Solubility and permeation of CO₂/CH₄ in ILs/PVDF hybrid membranes with an initial feed pressure of 10 bar and temperature of 298 K, (a–b) [Bmim][PF₆]/PVDF membrane, (c–d) [Bmim][NTf₂]/PVDF membrane. Left: membrane, right: permeate-side.

Table 2
Selectivity of CO₂/CH₄ in different hybrid membranes.

Selectivity	[Bmim][PF ₆]/PVDF membrane	[Bmim][NTf ₂]/PVDF membrane
Permselectivity	2.24	2.93
Solubility selectivity	2.15	2.74
Diffusion selectivity	1.04	1.07

between the hybrid membrane and CO₂ is stronger than that between the hybrid membrane and CH₄, which explains why the ILs/PVDF hybrid membrane has high CO₂ selectivity in separating CO₂/CH₄. In addition, the coordination numbers of CO₂ are 1.42 for [NTf₂][−] and 1.05 for [PF₆][−]. Meanwhile, the coordination number for CH₄ around [NTf₂][−] is also higher than that around [PF₆][−]. The results suggest that the distribution of CO₂/CH₄ around [NTf₂][−] are higher than those around [PF₆][−]. Moreover, it also can be seen that the coordination numbers for CO₂/CH₄ around anions are higher than those around PVDF, indicating that CO₂/CH₄ are mainly distribute around ILs in the hybrid membrane.

Furthermore, the interaction energies (van der Waals + Coulombic) of gas-[PF₆][−]/[NTf₂][−] and gas-PVDF were calculated and displayed in Fig. 8. It is evident that the interaction energies of CO₂-[PF₆][−]/[NTf₂][−] and CO₂-PVDF are stronger than those of CH₄-

[PF₆][−]/[NTf₂][−] and CH₄-PVDF, which leads to the rapid penetration of CO₂ into the membrane interior. Meanwhile, the interaction energies of gas-[NTf₂][−] are clearly stronger than those of gas-[PF₆][−]. Therefore, the results further validate that [Bmim][NTf₂]/PVDF hybrid membrane has better gas separation performance in separating CO₂/CH₄ than [Bmim][PF₆]/PVDF hybrid membrane.

3.5. Diffusion of cations and anions

Since CO₂/CH₄ are chiefly distribute around ILs, it is necessary to study the transport properties of ILs in hybrid membrane. The MSD of cations and anions for the studied hybrid membrane were calculated and shown in Fig. 9. The diffusion coefficient was gotten from the slop of MSD using Einstein's relation. The self-diffusion coefficients of cations and anions in [Bmim][PF₆]/PVDF membrane are 7.54×10^{-11} and 5.31×10^{-11} m²/s, respectively. While, the self-diffusion coefficients of cations and anions in [Bmim][NTf₂]/PVDF membrane are 9.98×10^{-11} and 7.95×10^{-11} m²/s. It can be seen that the self-diffusion coefficient of ions in [Bmim][PF₆]/PVDF hybrid membrane are smaller than those in [Bmim][NTf₂]/PVDF hybrid membrane. Since [Bmim][NTf₂] moves faster than [Bmim][PF₆] in the hybrid membrane, it can be inferred that the CO₂/CH₄ diffuse faster in the [Bmim][NTf₂]/PVDF hybrid membrane than those in [Bmim][PF₆]/PVDF hybrid membrane.

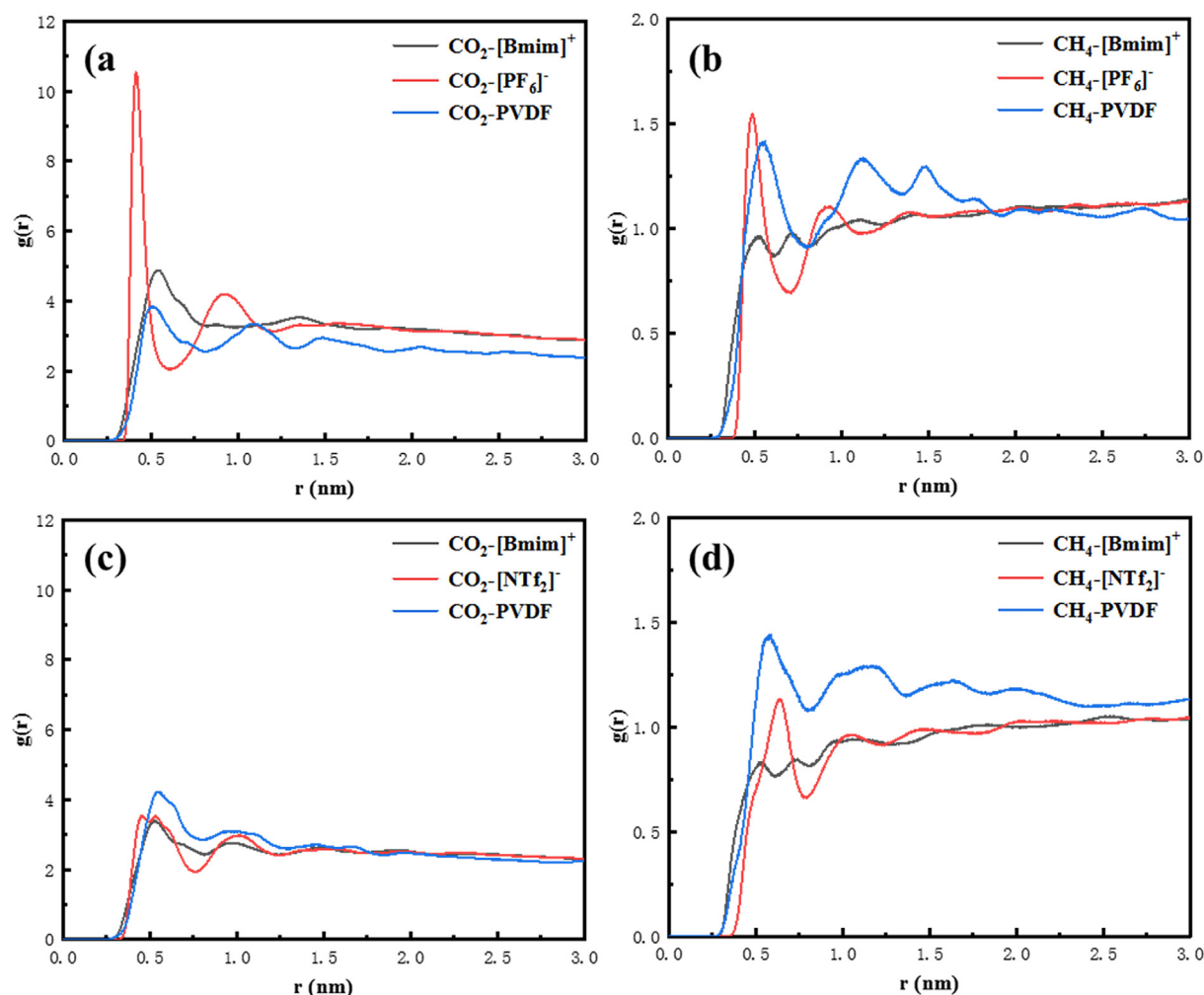


Fig. 7. The center-of-mass RDFs for gas-cations, gas-anions and gas-PVDF in (a–b) [Bmim][PF₆]/PVDF membranes and (c–d) [Bmim][NTf₂]/PVDF membranes.

Table 3

Coordination numbers of gas-[PF₆][−]/[NTf₂][−] and gas-PVDF in all ILs/PVDF hybrid membrane.

Coordination number	[Bmim][PF ₆]/PVDF membrane		[Bmim][NTf ₂]/PVDF membrane	
	CO ₂	CH ₄	CO ₂	CH ₄
anion	1.05	0.38	1.42	0.43
PVDF	0.88	0.29	0.86	0.30

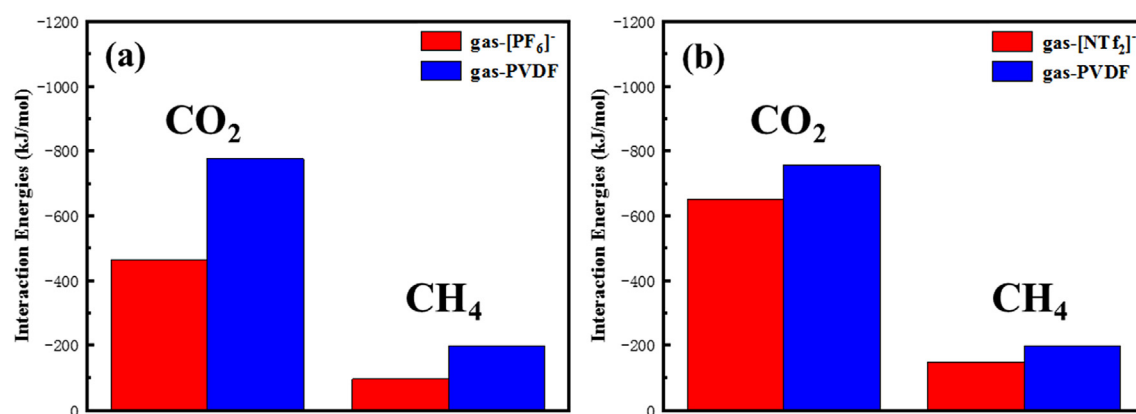


Fig. 8. Interaction energy of gas molecules with each component in (a) [Bmim][PF₆]/PVDF membrane and (b) [Bmim][NTf₂]/PVDF membrane.

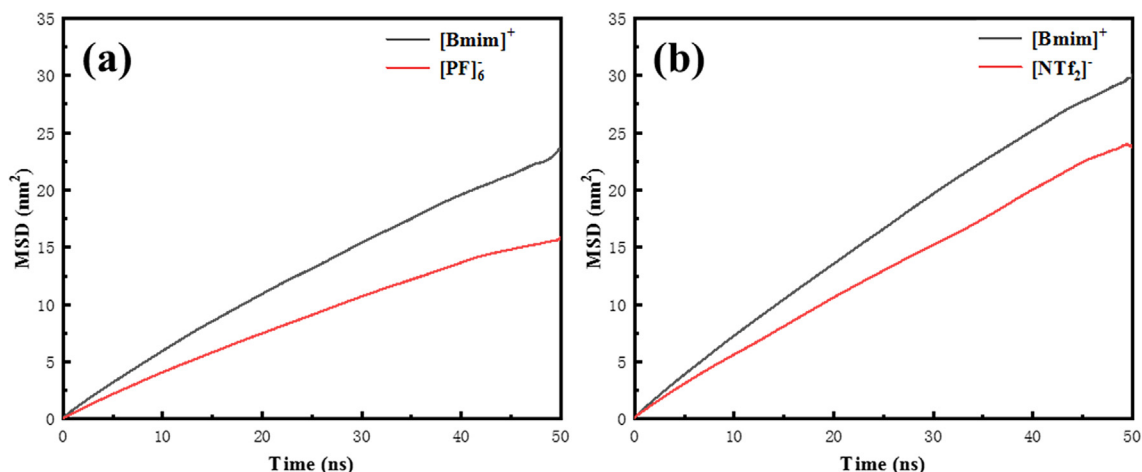


Fig. 9. MSD vs. time plots for (a) [Bmim][PF₆] and (b) [Bmim][NTf₂] in the hybrid membrane.

3.6. Transport process of CO₂

Since CO₂ is more permeable, the CO₂ transfer process were further studied. To quantitatively analyze how CO₂ penetrate from gas phase into the hybrid membrane, the potential of mean force (PMF) was obtained by umbrella sampling simulations [60]. As shown in Fig. 10a, for the system of CO₂-[Bmim][PF₆]/PVDF, starting from the gas phase, the PMF gradually decreases (more negatively) when moves toward the gas-membrane interface due to the strong interaction of CO₂ with the hybrid membrane. At the interface, the PMF is about $-4.11 \text{ kJ}\cdot\text{mol}^{-1}$, which corresponds to the driving force for CO₂ penetrating through the interface of hybrid membrane. Combination with the number density of the hybrid membrane in Fig. 5b, it can be found that CO₂ then gradually enter the main PVDF phase ($z \approx -3 \sim -1 \text{ nm}$) with an average PMF of $-12 \text{ kJ}\cdot\text{mol}^{-1}$. After that, CO₂ enter the main IL phase at $z \approx -1 \text{ nm}$ with an average PMF of $-27 \text{ kJ}\cdot\text{mol}^{-1}$. The PMF value of CO₂ in the main ILs phase is smaller than the PMF value of CO₂ entering the membrane interface, which indicates that the ILs enhance the solubility of CO₂. Whereas, for the system of CH₄-[Bmim][PF₆]/PVDF, the energy barrier is increased, which inhibits CH₄ from penetrating through the interface. In contrast, the CO₂ experiences a dropped energy barrier. Therefore, com-

pared with CH₄, CO₂ is easier to pass through the membrane. Moreover, the very small value of PMF corresponds to the dense CO₂ (Fig. 10b), which is because CO₂ molecules are more preferentially located at the minimal values of PMF.

In order to further reveal the transport process of CO₂, the angular probability density distribution of CO₂ was calculated and presented in Fig. 11. Since CO₂ is quickly attracted to the hybrid membrane interface, only the first 1.2 ns was analyzed. It is very interesting that the CO₂ is uniformly distributed in the gas layer in the initial configuration. However, as time increases, the angle of most of CO₂ at the interface gradually is approached 90°, which means that most of CO₂ are adsorbed to the interface by nearly paralleling to the membrane. This may be due to the large amount of PVDF distributed at the interface and the strong interaction between CO₂ and fluorine atoms on PVDF.

3.7. Effect of thickness

It is well known that the thickness of membrane has a significant effect on the gas separation. Thus, the permeation selectivity of CO₂/CH₄ in two ILs/PVDF hybrid membranes with different thicknesses were analyzed by calculating the average number of gas molecules permeating the vacuum side in the last 20 ns. The

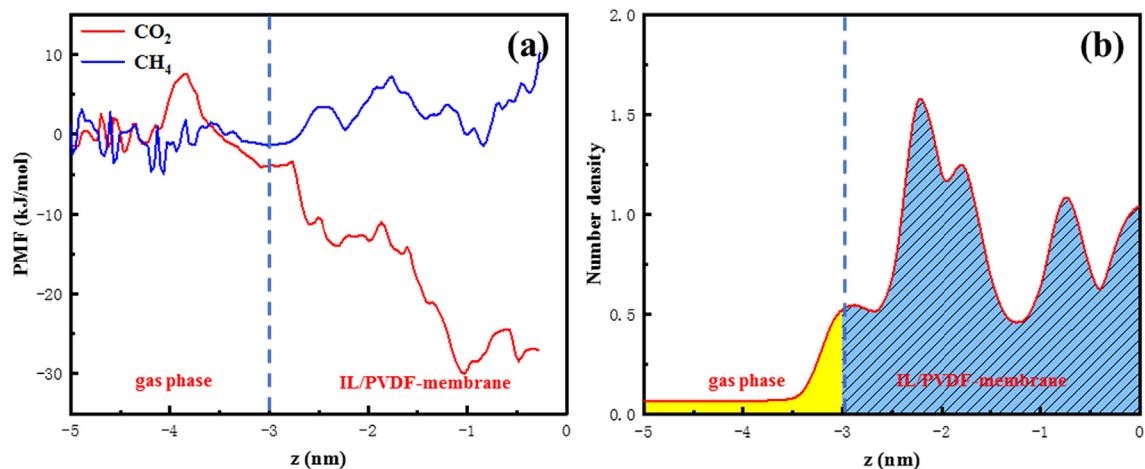


Fig. 10. (a) The potential of mean force of CO₂ and CH₄ penetrate from the gas phase into the hybrid membrane. (b) The number density profile of CO₂. The dashed line indicates the gas-membrane interface.

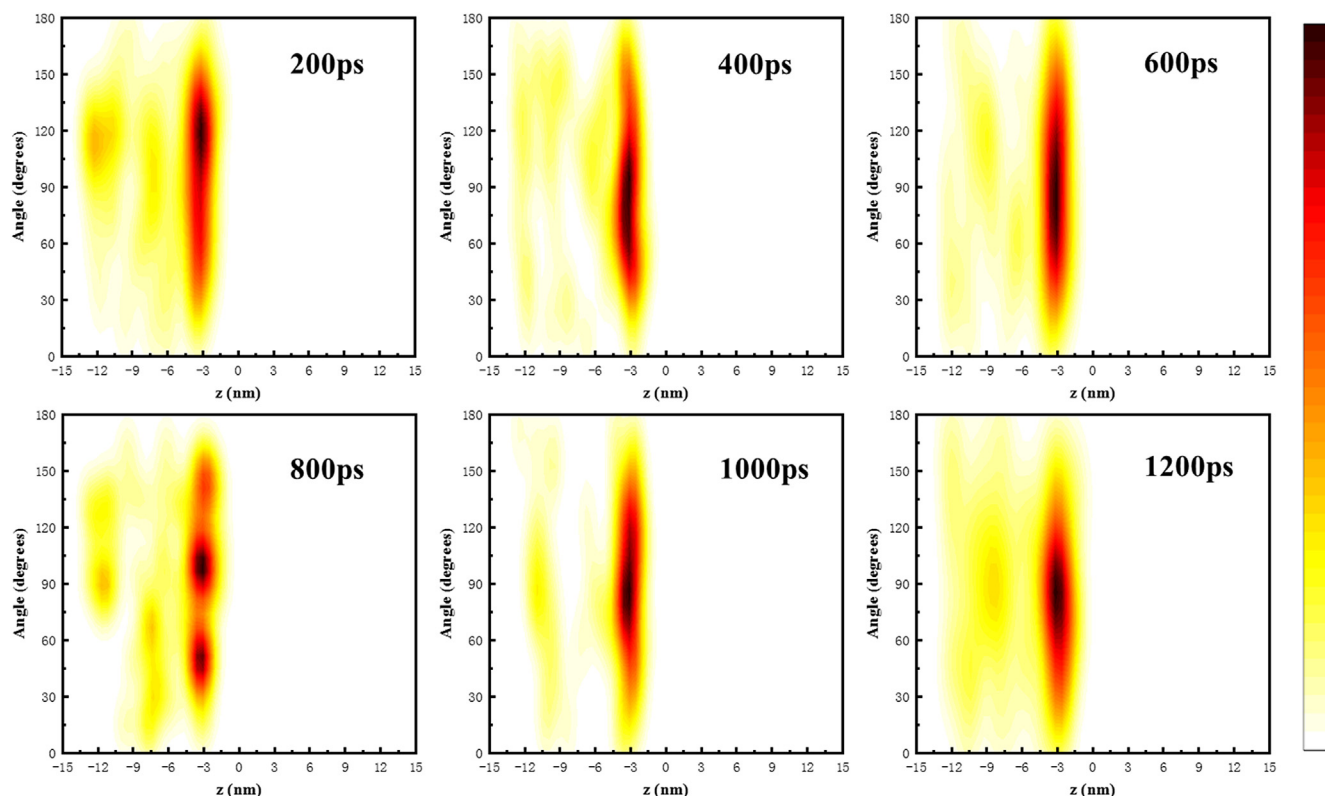


Fig. 11. The angular probability density distribution of CO₂. The angle of 90° indicates that the CO₂ are parallel to the interface of the membrane. $z = 10$ nm is the interface between the gas layer and the hybrid membrane. The darker color indicates the higher probability. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

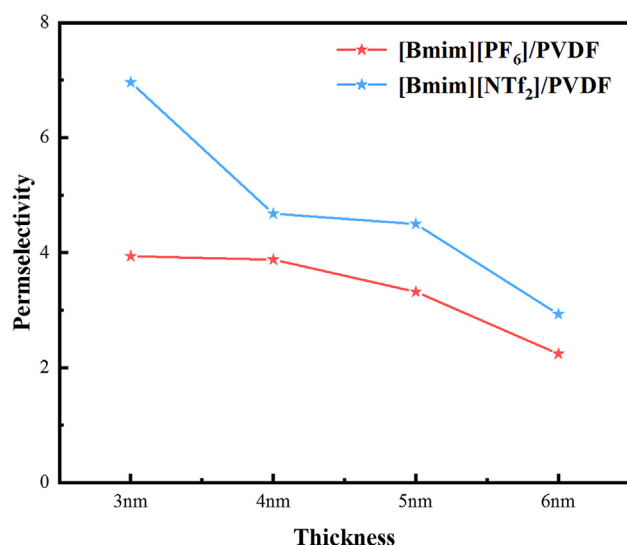


Fig. 12. Selectivity of CO₂/CH₄ in two ILs/PVDF hybrid membranes with different thicknesses.

thickness of 3, 4, 5 and 6 nm were investigated in our work, which can be considered to be ultra-thin membrane. The results are presented in Fig. 12, the permselectivity of CO₂/CH₄ in [Bmim][NTf₂]/PVDF membrane is 2.93 and 6.96 for 6 nm and 3 nm thicknesses, respectively. The permselectivity of CO₂/CH₄ is enhanced by more than twice at 3 nm. Therefore, for ultra thin membranes, the permselectivity would increase with the decrease of the thick-

ness. Similar findings have been reported experimentally. Guo et al. [61] prepared the ILs/CMS composite membranes and tested the gas separation by adjusting the ILs thickness. They found that the increase of the thickness of ILs selective layer on the membrane lead to the decrease of the permeability and CO₂/N₂ selectivity. Selyanchyn et al. [62] prepared Pebax-1657/PDMS composite membrane. It was reported that when the thickness of Pebax-1657 selective layer from 0.29 μ m to 0.39 μ m, the CO₂ permeance decreased from 300 GPU to 160 GPU, and the CO₂/N₂ selectivity decreased from 43 to 35, respectively. Again, Li et al. [63] found that CO₂/N₂ selectivity decreased with the increase in the thickness of PDMS layer from 3.3 to 5.5 μ m. Moreover, the permselectivity of CO₂/CH₄ in [Bmim][NTf₂]/PVDF membrane is higher than that in [Bmim][PF₆]/PVDF membrane for all the thicknesses.

As discussed above, the solubility selectivity plays a key role in CO₂/CH₄ separation. Thus, the center-of-mass RDFs between PVDF and [PF₆]⁻/[NTf₂]⁻ in two ILs/PVDF hybrid membranes with different thicknesses were analyzed. As shown in Fig. 13(a,b), the peak intensity of the RDFs of PVDF-[PF₆]⁻/[NTf₂]⁻ increases as the membrane thickness decreases for [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF system, indicating that the interaction between PVDF and anions increases with the decrease of membrane thickness. In addition, Fig. 13(c,d) shows the RDFs of CO₂-[PF₆]⁻/[NTf₂]⁻ with different membrane thickness in [Bmim][PF₆]/PVDF and [Bmim][NTf₂]/PVDF system. It can be found that the interaction between CO₂ and anions increases as the membrane thickness decreases. Therefore, it can be inferred that the strong interaction between PVDF and anions can promote the absorption and solubility of CO₂, and reducing the thickness of the ILs/PVDF membrane can increase the selectivity of CO₂/CH₄.

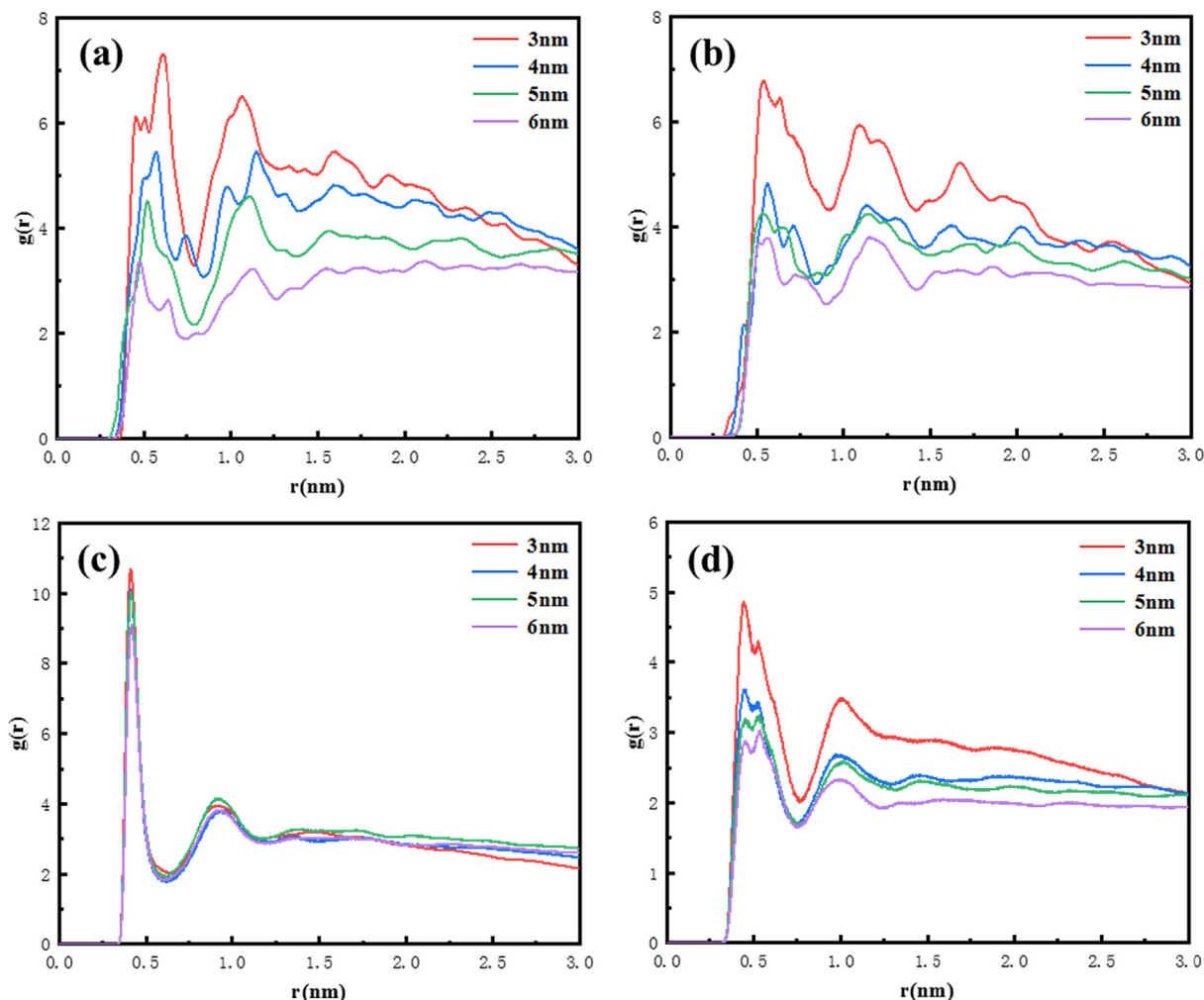


Fig. 13. The center-of-mass RDFs between (a) PVDF- $[\text{PF}_6]^-/[\text{NTf}_2]^-$ and (b) CO_2 - $[\text{PF}_6]^-/[\text{NTf}_2]^-$ with different membrane thickness in (left) $[\text{Bmim}][\text{PF}_6]/\text{PVDF}$ and (right) $[\text{Bmim}][\text{NTf}_2]/\text{PVDF}$ system.

4. Conclusions

In this paper, the separation process of CO_2/CH_4 in ILs/PVDF hybrid membranes ($[\text{Bmim}][\text{PF}_6]/\text{PVDF}$ and $[\text{Bmim}][\text{NTf}_2]/\text{PVDF}$) were investigated using molecular dynamics simulations. From the results of MSD and RDFs, it was found that the ILs could decrease the crystallinity of PVDF and improve the flexibility of PVDF chains, which is favor for the gas permeation and diffusion. The number densities show that the PVDF mainly distribute around the surface of both sides of the hybrid membrane, and the ILs are wrapped by PVDF, which could effectively avoid the loss of ILs. The results of permeation selectivity, solubility selectivity and diffusion selectivity demonstrated that compared with the diffusion selectivity, the solubility selectivity plays a dominant role in the separation of CO_2/CH_4 . In addition, the permeation selectivity, solubility selectivity, and diffusion selectivity of CO_2/CH_4 in $[\text{Bmim}][\text{NTf}_2]/\text{PVDF}$ membrane are higher than those in $[\text{Bmim}][\text{PF}_6]/\text{PVDF}$ membrane, confirming that $[\text{Bmim}][\text{NTf}_2]/\text{PVDF}$ membrane has better performance than $[\text{Bmim}][\text{PF}_6]/\text{PVDF}$ membrane in separating CO_2/CH_4 . Based on the RDFs and interaction energy between gas and ILs/PVDF, it was found that the interactions between CO_2 and ILs/PVDF are significantly stronger than those between CH_4 and ILs/PVDF, resulting in the rapid penetration of CO_2 into the membrane and the high selectivity of CO_2/CH_4 . In addition, the PMF for CO_2 at the gas–solid interface and the main IL phase is about $-4.11 \text{ kJ}\cdot\text{mol}^{-1}$ and $-27 \text{ kJ}\cdot\text{mol}^{-1}$, indicating that

CO_2 enters the ILs phase more easily. It also found that most of the CO_2 are adsorbed to the interface by nearly paralleling to the membrane by studying the angular probability density distribution of CO_2 . Moreover, the results of permselectivity of CO_2/CH_4 and RDFs of PVDF-anions and CO_2 -anions with different thicknesses (3, 4, 5 and 6 nm) reveal that the thinner the hybrid membrane, the stronger the interaction between the anion and PVDF, leading to the higher the selectivity of CO_2/CH_4 . This work provides valuable information for understanding the separation processes of CO_2/CH_4 in the ILs/PVDF hybrid membrane, and also provides theoretical guidance for designing new ILs/polymer hybrid membranes. However, due to the limitation of computational resources, we can only study nanoscale membranes at present. As membranes with micron thickness is indispensable for experiment, micron-scale membranes (e.g., using coarse-grained models) will be the further investigated in our future work, which will be more useful to provide guidance for fabricating high-performance ILs/polymer hybrid membranes.

CRediT authorship contribution statement

Zhenlei Wang: Investigation, Writing – original draft. **Fang Yan:** Investigation, Validation. **Lu Bai:** Investigation, Validation. **Xiangping Zhang:** Investigation, Validation. **Xiaomin Liu:** Supervision. **Xiaochun Zhang:** Supervision, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2022.119119>.

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